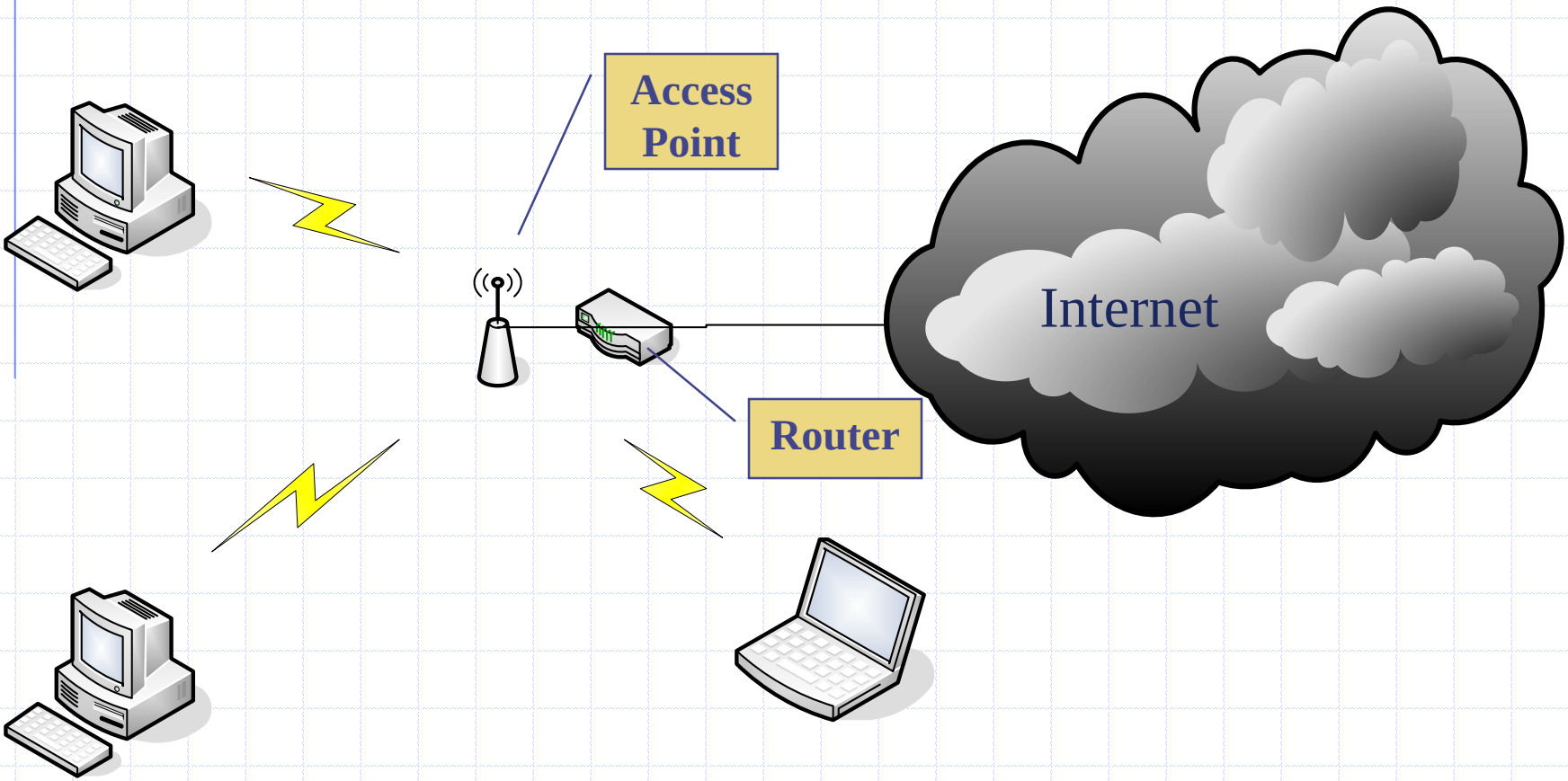
Wireless LANS

Motivation for Wireless LANs

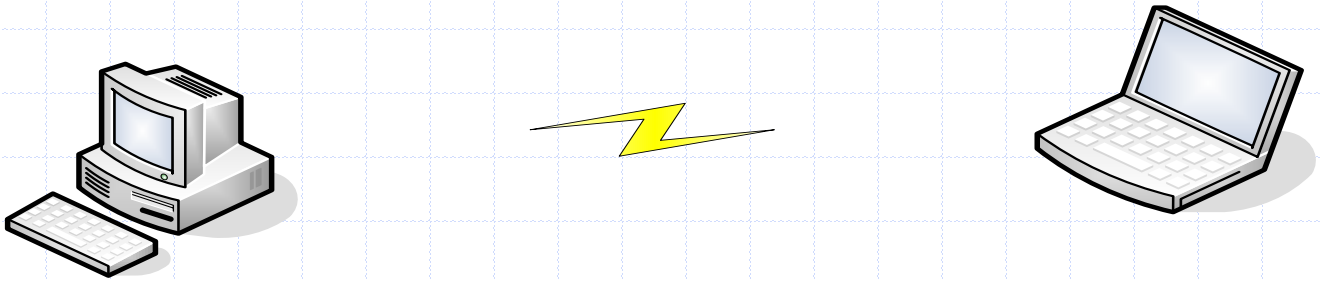
- Convenience
- Mobility
- Productivity
- Deployment
- Expandability
- Cost

Example Wireless Net



Example Wireless Net

Ad Hoc
(Point to Point)



Wireless Transmission Technologies

- Infrared LANs
- Narrowband Microwave LANs
- Spread Spectrum LANs

Infrared LANs

- Uses infrared light to carry the information
  - Very high potential bandwidth
  - Cheap
  - Unregulated
- Communications are limited to a single room
  - Allows reuse of spectrum in adjacent rooms
  - Limits eavesdropping

Infrared LANs

- Transmission Techniques
  - Directed beam infrared: Uses a focused IT data link. Can support a range up to a few kilometers under good conditions.
  - Omni-directional: Within a room, transmitter is placed in ceiling
  - Diffuse transmission: Unfocused signal is pointed to a reflecting surface (ceiling, walls) which then reflects to other receivers.

Narrowband Microwave

- Microwave transmission with limited bandwidth
- Licensed Narrowband RF
  - Typically in 18 GHz range
  - License area typically 28 km radius
- Unlicensed Narrowband RF
  - RadioLAN was first product
  - 5.8 GHz 10 Mbps transmission

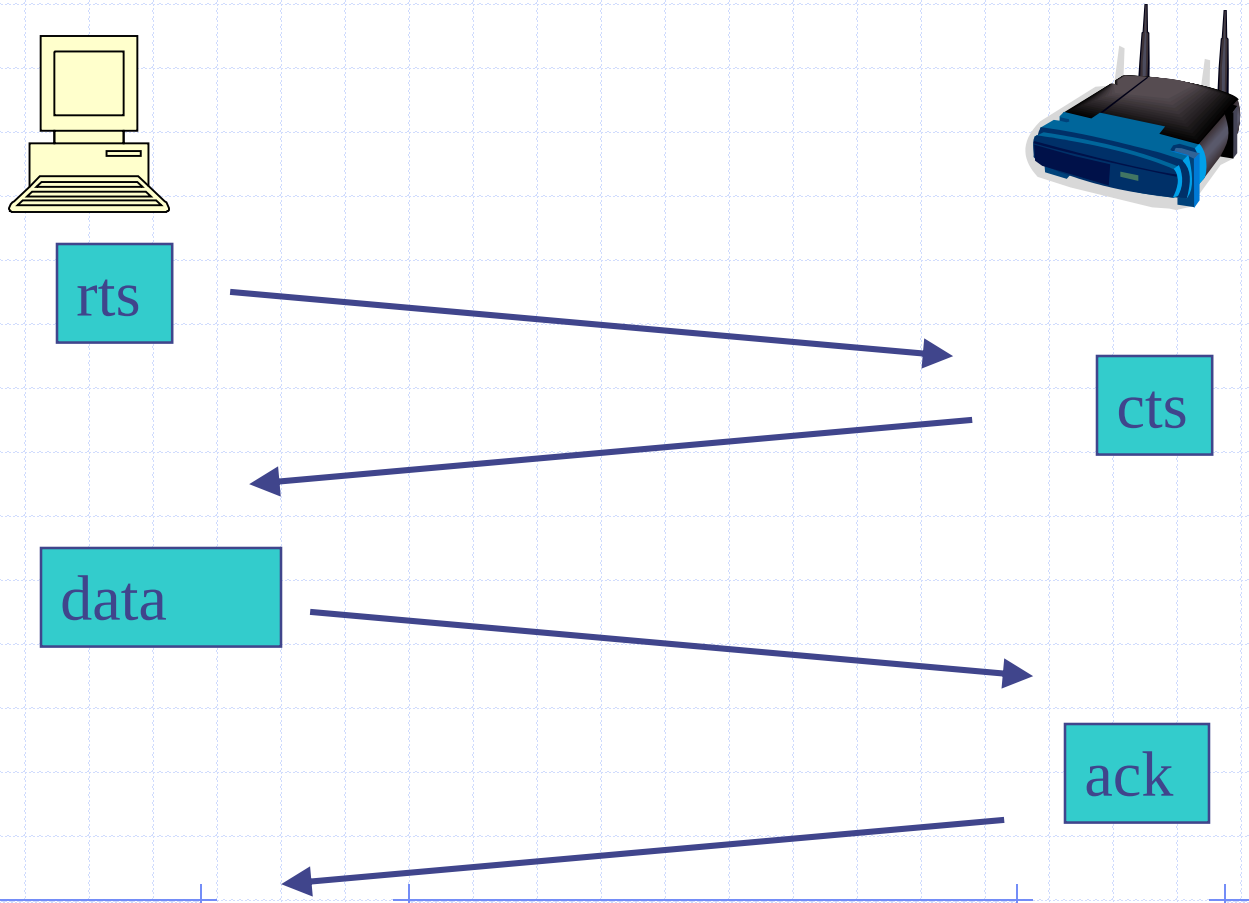
Spread Spectrum LANs

- Most common technique used today
  - DSSS (Direct sequence spread spectrum)
  - FHSS (Frequency hopping Spread Spectrum)

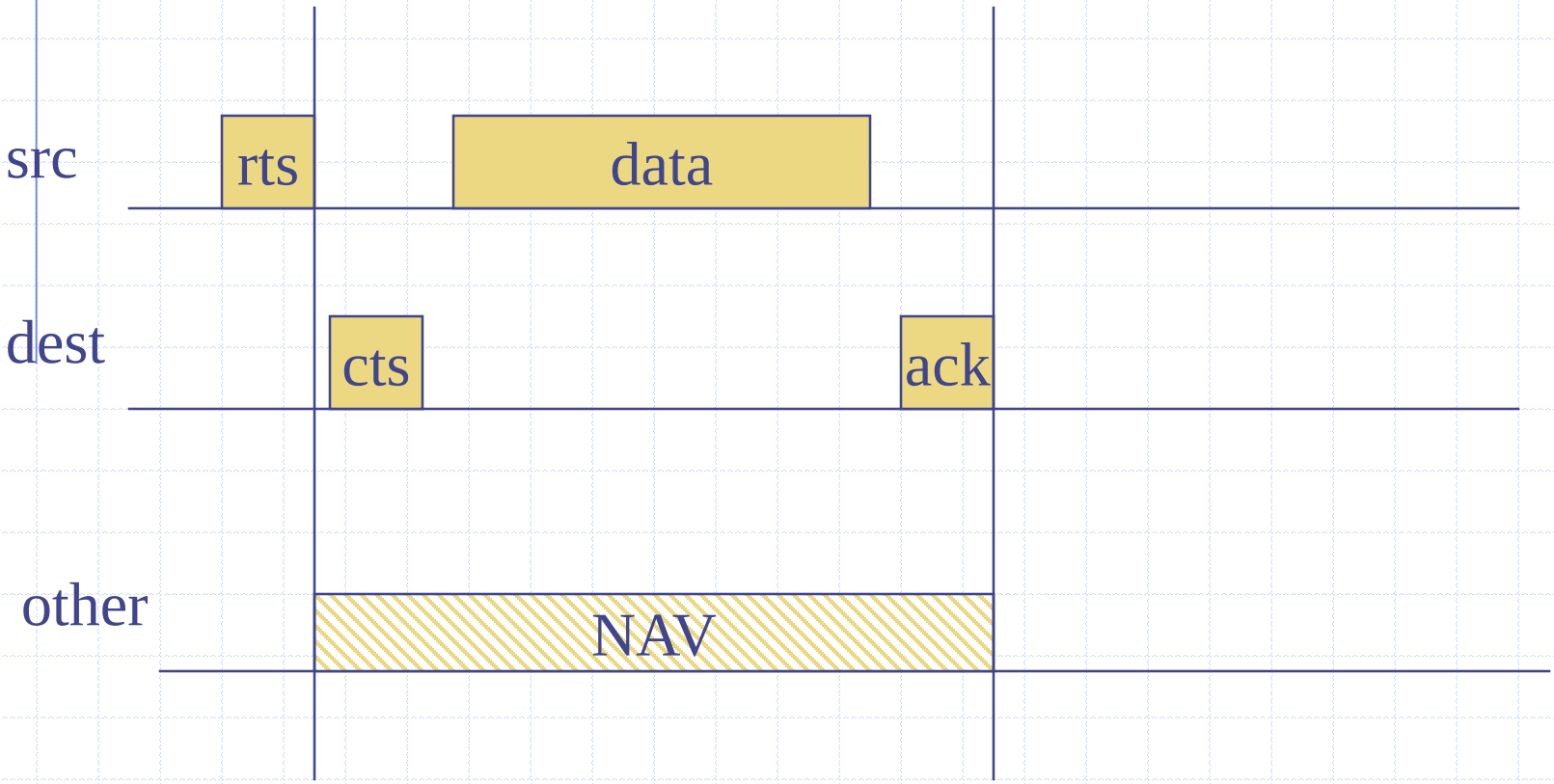
Wireless Comparison

	Infrared	Spread Spectrum	Microwave Radio
Data Rate (mbps)	1 to 10	2 to 54	10 to 20
Range (m)	15 to 60	30 to 250	10 to 40
Access	CSMA, +	CSMA	CSMA, +
License?	No	No	Yes

802.11 Communications



802.11 Communications



NAV: Network Allocation Vector

Summary

- Wireless LANs becoming much more common. Could support net access any time, anywhere.
- Several different wireless standards available today with different capabilities
- Wireless access brings with it new security and reliability issues compared to wired networks